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MIS 401: Business Intelligence & Analytics

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Diabetes Risk Prediction by Patient Health Indicators

UCI x CDC

MIS 401 Meeting Q&A Overview - Step 2 (9 am - 9:15 am)

Data Source

UCI Machine Learning Repository: CDC Diabetes Health Indicators

<https://archive.ics.uci.edu/dataset/891/cdc+diabetes+health+indicators>

Kaggle Dataset Homepage

<https://www.kaggle.com/datasets/alexteboul/diabetes-health-indicators-dataset>

The dataset is the CDC Diabetes Health Indicators dataset used in the proposal. It contains 253,680 observations, 21 predictor variables, and no missing values.

Response Variable

Diabetes_012

- 0 = no diabetes
- 1 = prediabetes
- 2 = diabetes

Variable Types

Categorical Predictors

- HighBP
- HighChol
- CholCheck
- Smoker
- Stroke
- HeartDiseaseorAttack
- PhysActivity
- Fruits

- Veggies
- HvyAlcoholConsump
- AnyHealthcare
- NoDocbcCost
- DiffWalk
- Sex

Continuous / Ordinal Predictors

- BMI
- GenHlth
- MentHlth
- PhysHlth
- Age
- Education
- Income

Methods

- logistic regression
- k-nearest neighbors (KNN)
- decision tree

Response Variable Distribution

Counts

- No diabetes: 213,703
- Prediabetes: 4,631
- Diabetes: 35,346

Percentages

- No diabetes: 84.24%
- Prediabetes: 1.83%
- Diabetes: 13.93%

Preliminary Interpretation Notes

- The response variable is highly imbalanced, with most observations in the no diabetes group.

- BMI appears to increase with diabetes status.
- Age category appears to increase from no diabetes to diabetes.
- General health appears worse in the diabetes group.
- High blood pressure is much more common in the diabetes group than in the no diabetes group.
- High cholesterol is also more common in the diabetes group than in the no diabetes group.
- Physical activity appears lower in the diabetes group than in the no diabetes group.
- Sex shows some variation across diabetes status, but the relationship appears weaker than those for BMI, age, blood pressure, and cholesterol.

Stand-Out Notes

The strongest early patterns appear to be class imbalance, higher BMI, older age categories, worse general health, and higher rates of high blood pressure and high cholesterol in the diabetes group.

Visuals Included

Histograms

- BMI
- Age
- MentHlth
- PhysHlth

Boxplots

- BMI by Diabetes_012
- Age by Diabetes_012
- GenHlth by Diabetes_012
- MentHlth by Diabetes_012

Categorical Comparisons

- HighBP by Diabetes_012
- HighChol by Diabetes_012
- PhysActivity by Diabetes_012
- Sex by Diabetes_012

Diabetes + Pre-Diabetes

Relationship between each relationship and response variable: bar chart or pie charts (categorical) or histogram (continuous)

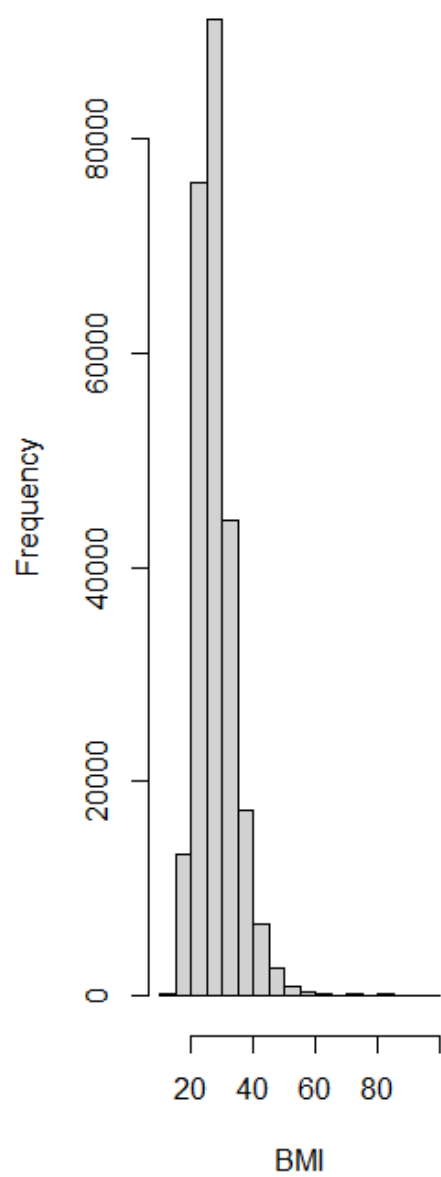
Relationship between each response variable and predictor: box plot for diabetic and nondiabetic (to see BMI positive or negative for example, Blood Pressure is higher or lower))

- Categorical Predictors : barplot()**

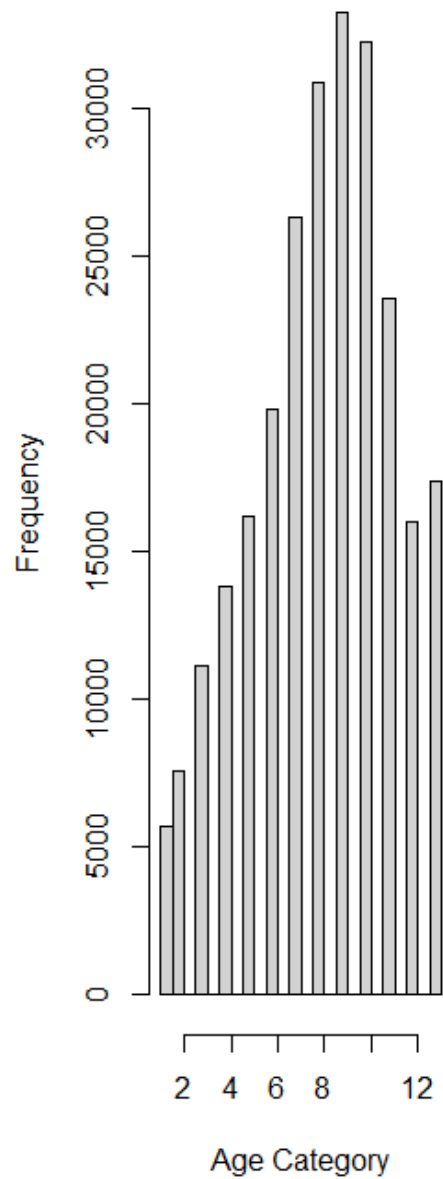
Section A: Histograms

1. Histogram of BMI
2. Histogram of Age Category
3. Histogram of Mental Health Days
4. Histogram of Physical Health Days

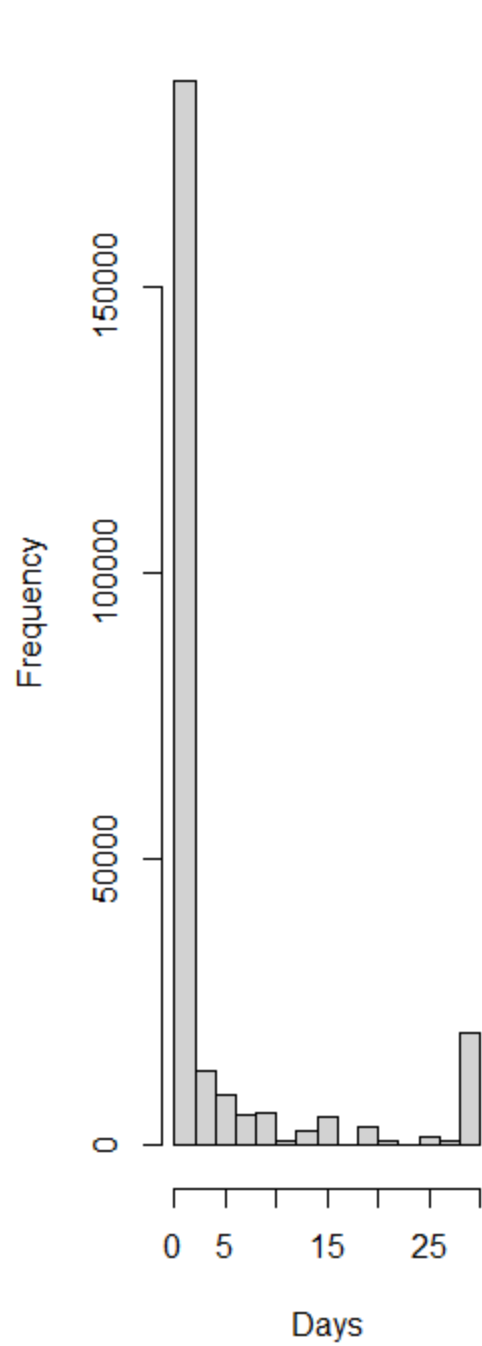
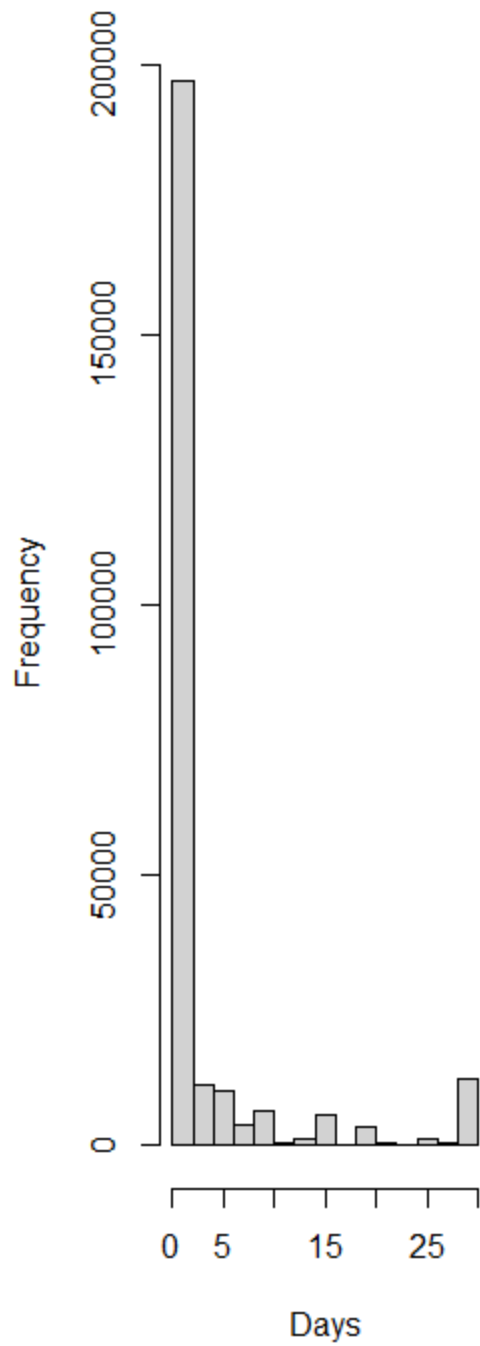
Histogram of BMI



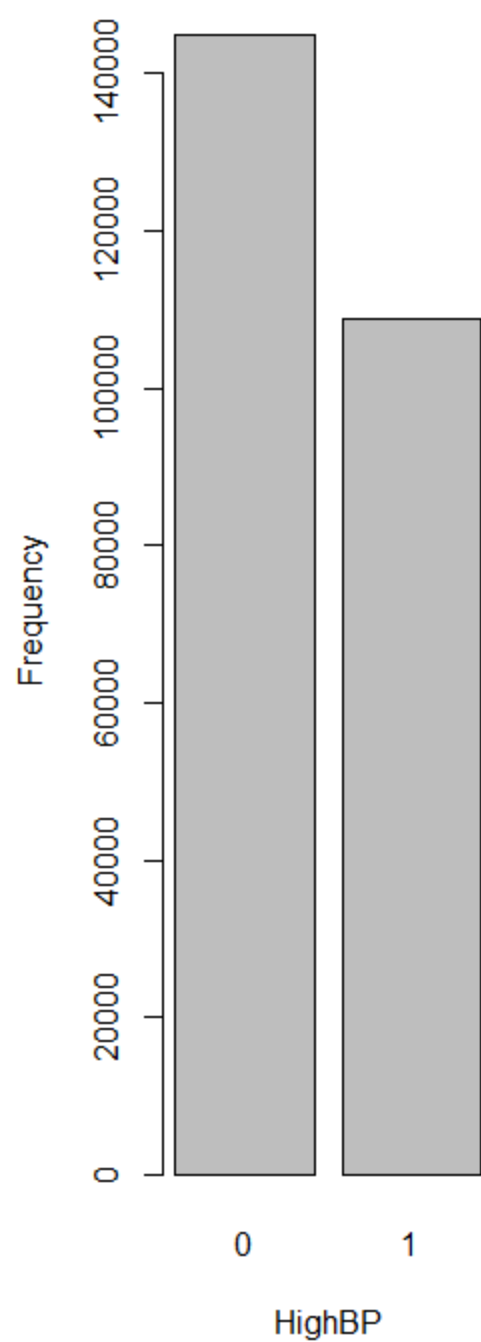
Histogram of Age Category



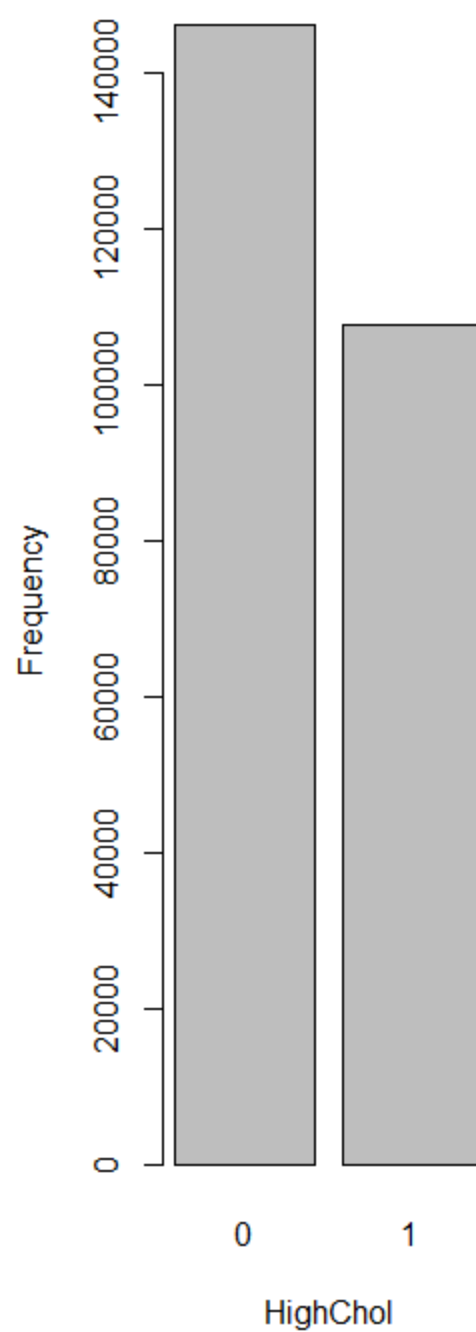
Histogram of Mental Health Data Histogram of Physical Health Data

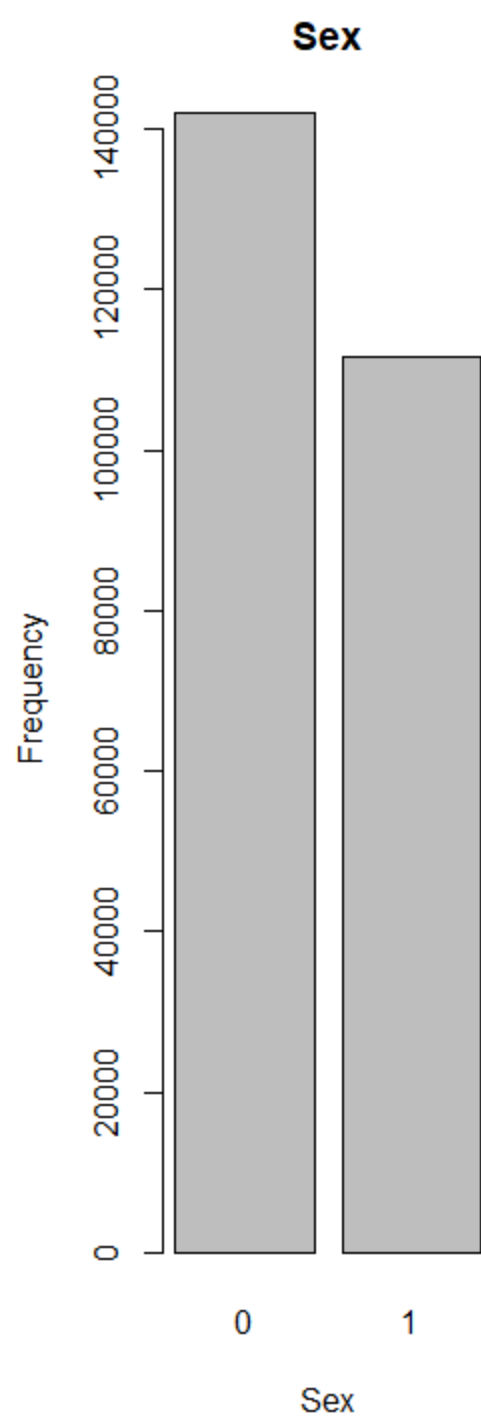
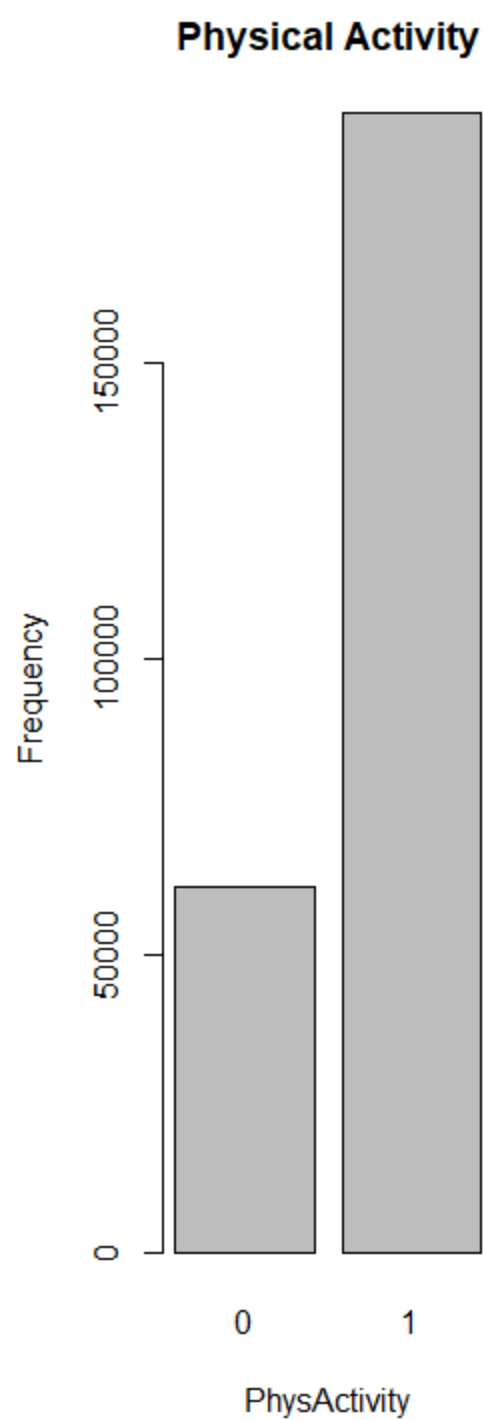


High Blood Pressure



High Cholesterol

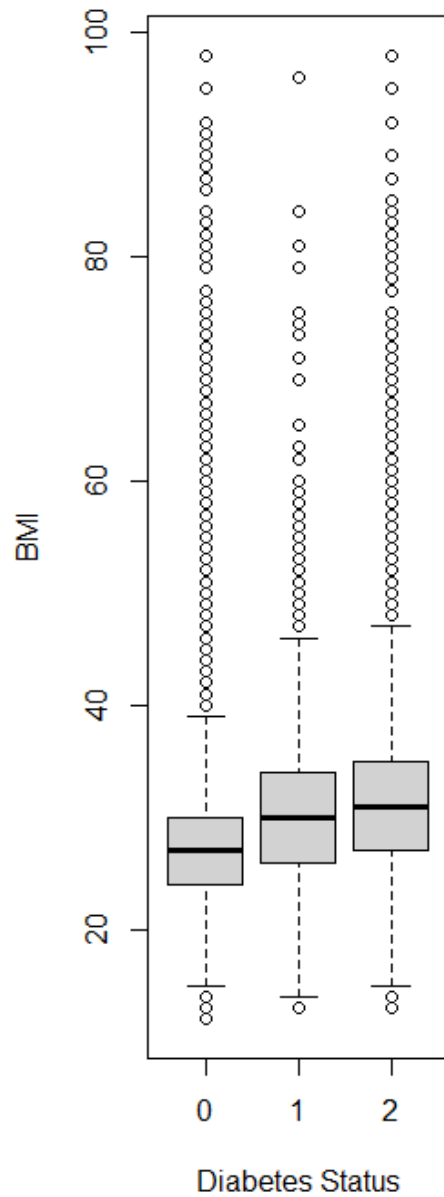




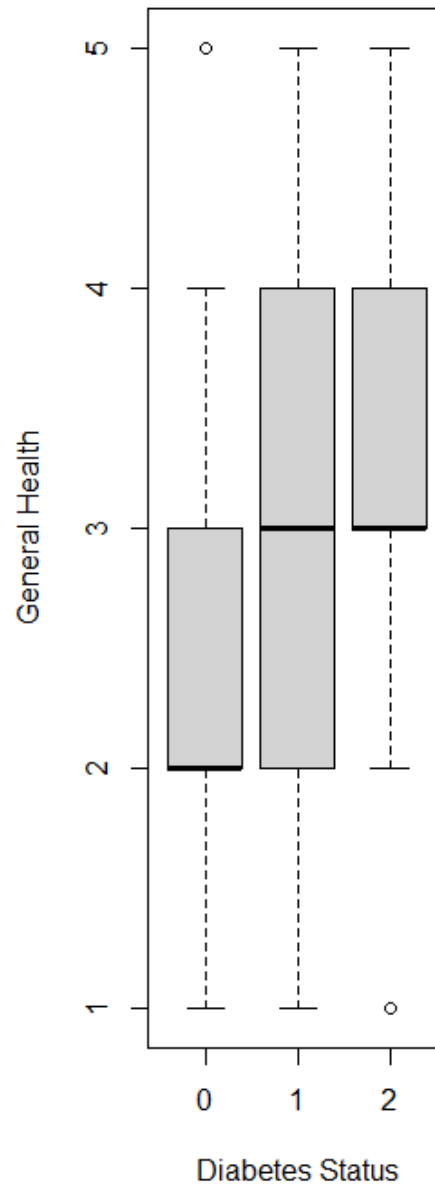
Section B: Boxplots by response

1. BMI by Diabetes Status
2. Age by Diabetes Status
3. General Health by Diabetes Status
4. Mental Health by Diabetes Status

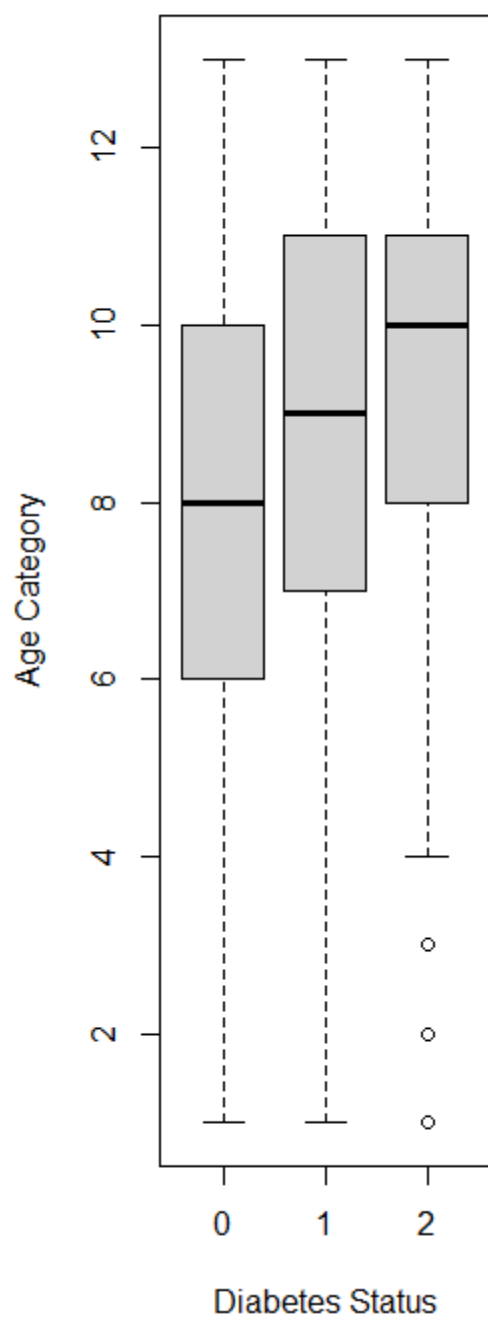
BMI by Diabetes Status



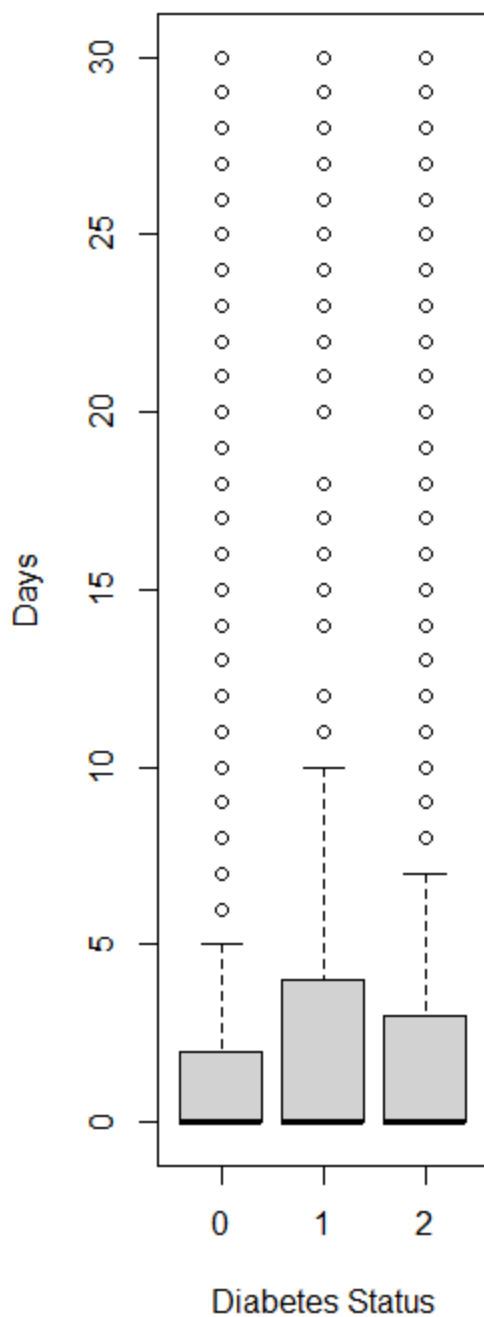
General Health by Diabetes Status



Age by Diabetes Status



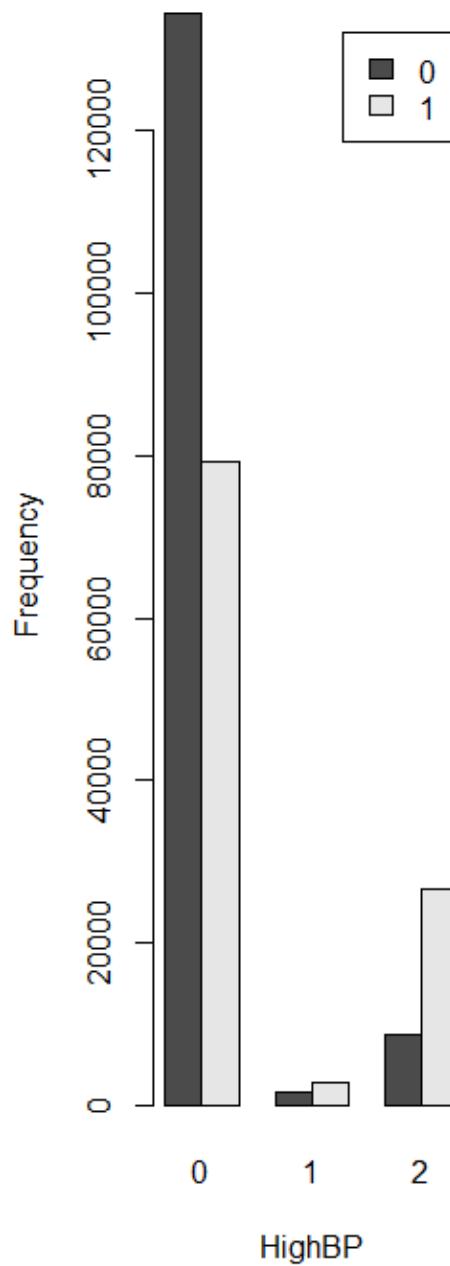
Mental Health by Diabetes Status



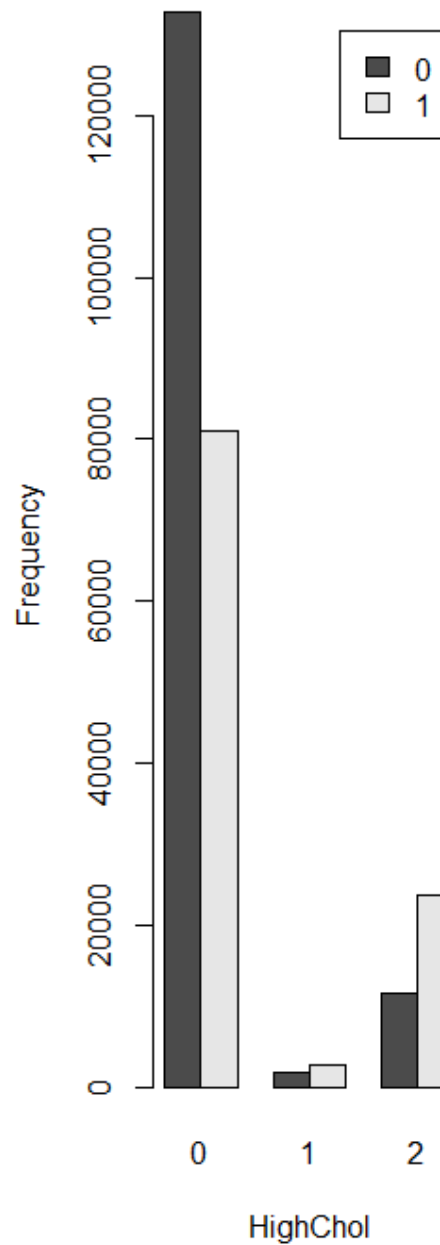
Section C: Categorical predictor comparisons

1. HighBP vs Diabetes Status
2. HighChol vs Diabetes Status
3. Physical Activity vs Diabetes Status
4. Sex vs Diabetes Status

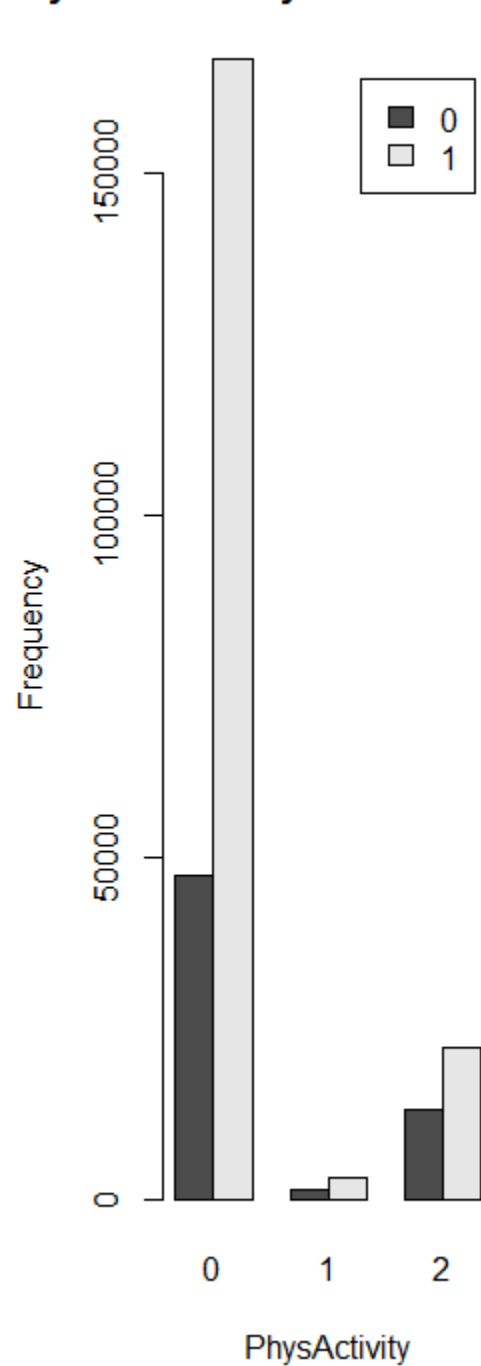
HighBP vs Diabetes Status



HighChol vs Diabetes Status



Physical Activity vs Diabetes St



Sex vs Diabetes Status

